

# Analysis for Applied Mathematics

Measure, function spaces, Fourier methods, stochastic tools, and variational ideas

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## Abstract

We develop the analytic toolkit most useful for modern applied mathematics: real analysis, measure and integration, functional analysis, Fourier and harmonic methods, variational ideas, stochastic analysis, and high-dimensional tools that support PDE, numerics, optimization, inverse problems, and learning.

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## 1 PART I — REAL ANALYSIS FOUNDATIONS

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### 1.1 1. Metric Spaces and Topology

- Metrics, norms, inner products; equivalence and comparison
- Open and closed sets; interior, closure, boundary
- Compactness: sequential, covering, Heine-Borel, and equivalence in metric spaces
- Completeness; Baire category theorem and its consequences
- Connectedness; path-connectedness

### 1.2 2. Sequences and Function Spaces

- Pointwise vs. uniform convergence; counterexamples
- Equicontinuity; Arzelà-Ascoli theorem
- Stone-Weierstrass theorem; polynomial approximation
- Power series; radius of convergence; real analytic functions

### 1.3 3. Multivariable Differentiation

- Fréchet derivative; directional derivatives; their relationship
- Chain rule; mean value inequality in  $\mathbb{R}^n$
- Inverse and implicit function theorems
- Higher derivatives; Taylor's theorem; local extrema and second-order conditions

### 1.4 4. Why a Better Integral is Needed

- Riemann-Stieltjes integral; limitations
- Failure modes of Riemann: pointwise limits, Fourier series
- Informal preview of measure theory motivation
- Key phenomena that force a better integral

## 2 PART II — MEASURE, INTEGRATION, AND PROBABILISTIC TOOLS

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### 2.1 5. Measures and $\sigma$ -Algebras

- $\sigma$ -algebras; measurable spaces; Borel sets
- Measures; outer measures; Carathéodory extension theorem
- Regularity properties; Borel and Radon measures on  $\mathbb{R}^n$

## 2.2 6. Lebesgue Integration

- Measurable functions; simple functions and approximation
- Lebesgue integral: definition through non-negative functions
- Monotone convergence; Fatou's lemma; dominated convergence
- Relation to Riemann; null sets and almost-everywhere statements

## 2.3 7. $L^p$ Spaces

- Hölder and Minkowski inequalities
- Completeness: Riesz-Fischer theorem
- Dense subsets: continuous and smooth approximation
- Dual of  $L^p$  ( $1 < p < \infty$ ); weak convergence in  $L^p$

## 2.4 8. Differentiation Theory

- Hardy-Littlewood maximal function; weak  $(1, 1)$  estimate
- Lebesgue differentiation theorem
- Absolutely continuous functions; Radon-Nikodym theorem
- Functions of bounded variation; Jordan decomposition

## 2.5 9. Product and Signed Measures

- Product  $\sigma$ -algebras and product measures
- Fubini and Tonelli theorems; interchange of integrals
- Hahn and Jordan decompositions
- Radon-Nikodym theorem and Lebesgue-Radon-Nikodym decomposition
- Conditional expectation as Radon-Nikodym derivative (preview)

## 2.6 10. Concentration Inequalities

- Markov and Chebyshev; moment generating functions
- Sub-Gaussian random variables; Hoeffding's inequality
- Bernstein's inequality; sub-exponential variables
- Azuma-Hoeffding for martingales; McDiarmid's bounded differences method
- Applications: generalization bounds in learning, MCMC mixing

## 2.7 11. Large Deviations

- Cramér's theorem on  $\mathbb{R}$ ; rate function; Legendre-Fenchel connection
- Gärtner-Ellis theorem; tilting
- Sanov's theorem; relative entropy as rate function
- Concentration as finite-sample control; large deviations as asymptotic control

## 3 PART III — FUNCTIONAL ANALYSIS AND FUNCTION SPACES

### 3.1 12. Normed Spaces and Banach Spaces

- Norms; completeness; examples ( $\ell^p$ ,  $L^p$ ,  $C(K)$ , Sobolev)
- Finite vs. infinite dimensions; Riesz lemma on non-compactness of unit ball
- Baire category theorem in Banach spaces; generic properties
- Series in Banach spaces; absolute vs. unconditional convergence

### 3.2 13. Bounded Linear Operators

- Operator norm; space  $B(X, Y)$  and its completeness
- Open mapping theorem and bounded inverse theorem
- Closed graph theorem
- Uniform boundedness principle (Banach-Steinhaus)

### 3.3 14. Hilbert Spaces

- Inner products; parallelogram law; polarization identity
- Orthonormal bases; Gram-Schmidt; Parseval's identity
- Projection theorem; orthogonal complements; best approximation
- Riesz representation theorem; self-duality of  $L^2$

### 3.4 15. Dual Spaces and Weak Topologies

- Hahn-Banach: analytic and geometric forms; separation
- Dual spaces of  $L^p$ ,  $\ell^p$ ,  $C(K)$
- Reflexive spaces; bidual embedding
- Weak and weak-\* convergence; Banach-Alaoglu theorem

### 3.5 16. Compact and Unbounded Operators

- Compact operators; precompact sets; examples
- Fredholm alternative; index theory (brief)
- Spectral theorem for compact self-adjoint operators
- Closed operators; adjoints; symmetric vs. self-adjoint
- $C_0$ -semigroups; Hille-Yosida theorem; evolution equations

### 3.6 17. Distributions

- Test functions  $D(\Omega)$ ; mollifiers; regularization
- Distributions as linear functionals; Dirac  $\delta$  done properly
- Differentiation and multiplication by smooth functions
- Fundamental solutions; PDE examples (Laplacian, heat)

### 3.7 18. Sobolev Spaces

- Weak derivatives;  $W^{k,p}$  and  $H^k = W^{k,2}$  spaces
- Sobolev embedding theorems (integer and fractional)
- Trace theorem; extension operators
- Poincaré inequality; Rellich-Kondrachov compactness

## 4 PART IV — FOURIER AND HARMONIC ANALYSIS

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### 4.1 19. Fourier Analysis on $\mathbb{R}^n$

- Fourier transform; Plancherel theorem; Hausdorff-Young
- Schwartz space  $S(\mathbb{R}^n)$  and tempered distributions
- Fourier transform and regularity:  $H^s$  Sobolev spaces
- Paley-Wiener theorem; analytic signals

### 4.2 20. Fourier Analysis on Groups

- Locally compact abelian groups; Haar measure; Pontryagin duality
- Fourier series on  $\mathbb{T}^n$ ; discrete Fourier transform on  $\mathbb{Z}/N\mathbb{Z}$
- Peter-Weyl theorem for compact groups
- Representation theory as generalized Fourier analysis

### 4.3 21. Singular Integrals and Calderón-Zygmund Theory

- Singular integrals: Hilbert transform, Riesz transforms
- Calderón-Zygmund decomposition; good/bad parts
- $L^p$  boundedness via weak-type  $(1, 1)$  and Marcinkiewicz interpolation
- $T(1)$  theorem (statement); connections to elliptic PDE regularity

### 4.4 22. Littlewood-Paley Theory and Wavelets

- Dyadic decomposition; Littlewood-Paley square functions
- Characterization of  $L^p$  and  $H^s$  norms via LP decomposition
- Multiresolution analysis; Daubechies wavelets; vanishing moments
- Fast wavelet transform; applications to signal processing and PDEs

### 4.5 23. Hardy Spaces and BMO

- Real Hardy space  $H^1$  and its atomic decomposition
- BMO: bounded mean oscillation; John-Nirenberg inequality
- Duality  $H^1 \leftrightarrow \text{BMO}$
- Endpoint estimates for singular integrals

### 4.6 24. Interpolation and Spectral Methods

- Riesz-Thorin complex interpolation
- Marcinkiewicz weak-type interpolation
- $K$ -functional and real interpolation (brief)
- Spectrum and resolvent; spectral radius formula
- Spectral theorem and Borel functional calculus
- Applications: wave propagation, Schrödinger equation

## 5 PART V — VARIATIONAL, CONVEX, AND TRANSPORT METHODS

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### 5.1 25. Variational Methods and Calculus of Variations

- Euler-Lagrange equations; derivation and examples
- Direct methods: lower semicontinuity, coercivity, existence

- Convex and polyconvex energies; relaxation
- Regularity; connection to PDEs

### 5.2 26. Convex Analysis

- Convex sets and functions; epigraph; closure
- Subdifferentials; Fenchel conjugate (Legendre-Fenchel transform)
- Minimax theorems; saddle points; von Neumann
- Variational inequalities; connection to optimization

### 5.3 27. Optimal Transport

- Monge problem; Kantorovich relaxation; linear programming formulation
- Wasserstein distances; metrization of weak convergence
- Brenier's theorem; polar factorization
- Wasserstein gradient flows; JKO minimizing movement scheme
- Connections to inverse problems, PDE, and generative modeling

## 6 PART VI — STOCHASTIC, HIGH-DIMENSIONAL, AND LEARNING-THEORETIC ANALYSIS

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### 6.1 28. Stochastic Analysis

- Brownian motion: construction, path properties, quadratic variation
- Martingales in continuous time; optional stopping; Doob's inequalities
- Itô integral and Itô's formula
- SDEs: existence and uniqueness; Girsanov; Feynman-Kac formula

### 6.2 29. Ergodic Theory

- Measure-preserving transformations; ergodicity; invariant measures
- Von Neumann  $L^2$  ergodic theorem
- Birkhoff pointwise ergodic theorem
- Mixing conditions; spectral characterization
- Applications: MCMC convergence; statistical mechanics; time series

### 6.3 30. Random Matrix Theory

- Wigner semicircle law: empirical spectral distribution of symmetric random matrices
- Marchenko-Pastur law: spectrum of sample covariance matrices
- Tracy-Widom distribution; universality (conceptual statement)
- Applications: PCA in high dimensions; deep learning initialization; free probability

### 6.4 31. Compressed Sensing and Sparse Recovery

- Sparse signals;  $\ell_1$ -relaxation; basis pursuit
- Restricted Isometry Property (RIP); incoherence
- RIP from Gaussian and Fourier random matrices
- Recovery guarantees; LASSO and statistical interpretation

### 6.5 32. Empirical Processes and Learning Theory

- Glivenko-Cantelli; uniform laws of large numbers
- Rademacher complexity; connections to covering numbers
- VC dimension and the VC inequality
- PAC learning; generalization bounds from empirical process theory

### 6.6 33. Neural Network Approximation Theory

- Universal approximation: Cybenko, Hornik
- Approximation rates for smooth functions; depth vs. width tradeoffs
- Barron spaces; integral representation of shallow networks
- Curse of dimensionality: what networks overcome and what they do not
- Connections to ridge functions and harmonic analysis

## 7 PART VII — ANALYTIC TOOLBOX FOR APPLIED MATHEMATICS

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### 7.1 34. Convolution, Kernels, and Smoothing

- Convolution as localization and smoothing
- Approximate identities and mollification

- Heat kernel intuition and regularization mechanisms
- Fourier-side interpretation of filtering

### **7.2 35. Compactness and Weak Convergence Methods**

- Arzelà-Ascoli, Rellich, and Prokhorov as compactness tools
- Weak convergence, weak-\* convergence, and tightness
- Passing to limits in nonlinear problems
- Existence proofs in PDE, probability, and variational problems

### **7.3 36. Energy, Duality, and Bootstrap Arguments**

- Energy estimates and a priori bounds
- Duality arguments and transference of estimates
- Bootstrap arguments for regularity improvement
- Perturbation methods and fixed-point methods

### **7.4 37. Reading Modern Applied Analysis**

- TT\* arguments and interpolation tricks
- Dyadic decompositions; soft vs. hard cutoffs
- When to use Sobolev, Fourier, variational, or probabilistic tools
- How modern papers combine analysis with numerics, optimization, and learning